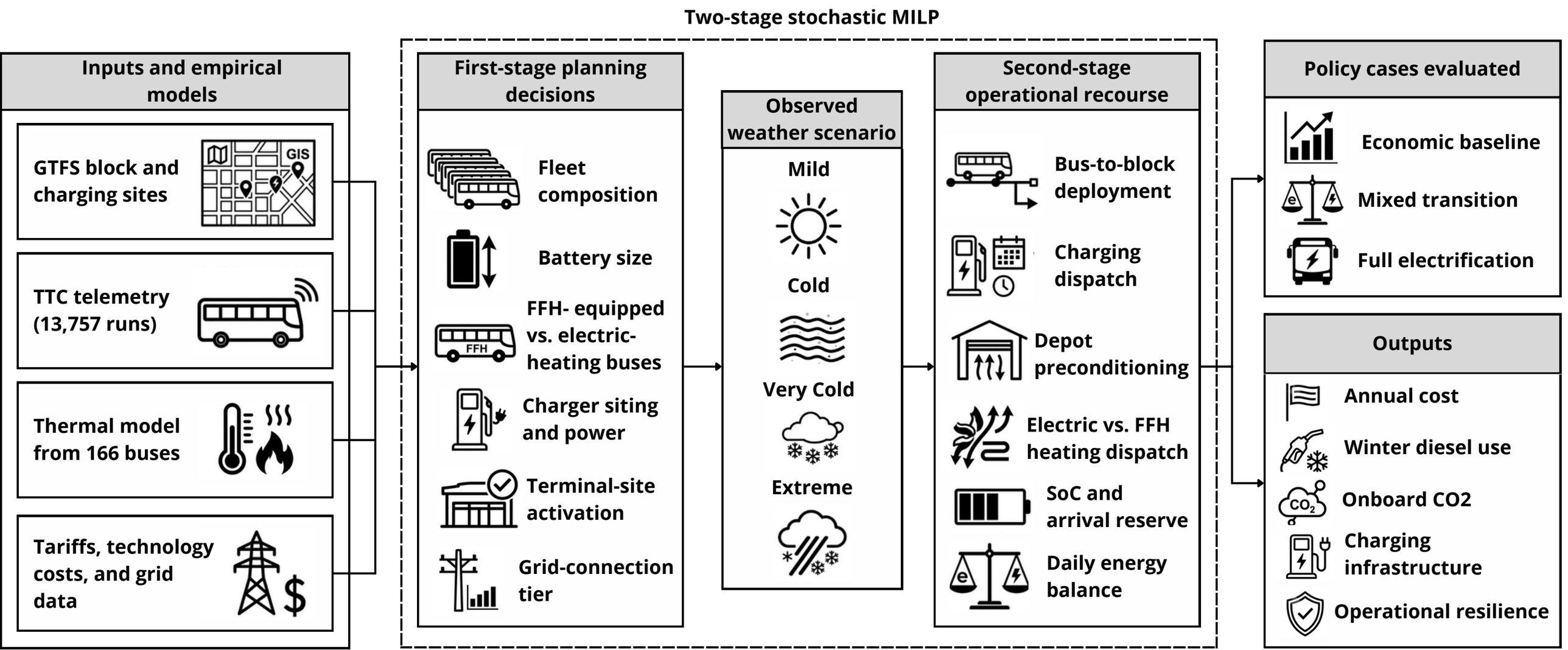

Two-stage stochastic MILP
Inputs and empirical models
GTFS block and charging sites
GIS
TTC telemetry (13,757 runs)
Thermal model from 166 buses
Tariffs, technology costs, and grid data
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First-stage planning decisions
Fleet composition
Battery size
FFH
FFH- equipped vs. electric-heating buses
Charger siting and power
Terminal-site activation
Grid-connection tier
Observed weather scenario
Mild
Cold
Very Cold
Extreme
Second-stage operational recourse
Bus-to-block deployment
Charging dispatch
Depot preconditioning
Electric vs. FFH heating dispatch
SoC and arrival reserve
Daily energy balance
Policy cases evaluated
Economic baseline
Mixed transition
Full electrification
Outputs
Annual cost
Winter diesel use
CO2
Onboard CO2
Charging infrastructure
Operational resilience